

NUCATS | Dean's Administration Annual Meeting

September 2nd, 2026

Welcome | Agenda

Purpose: This annual meeting brings together NUCATS leadership and the Dean's Administration to review progress toward strategic goals, discuss emerging opportunities and challenges, and ensure alignment between NUCATS initiatives and Feinberg's institutional priorities.

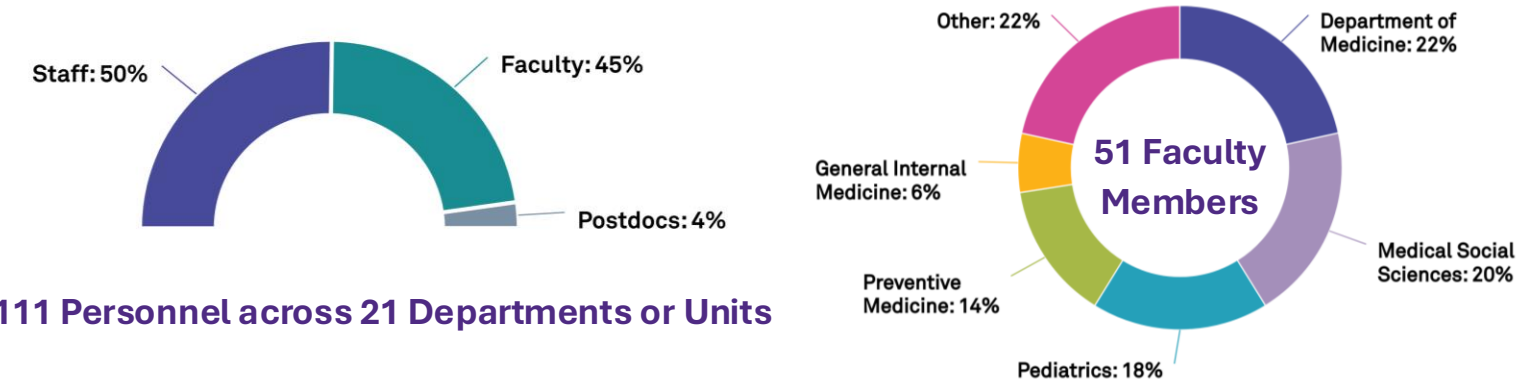
Agenda:

- I. NUCATS FY26 Highlights
- II. Informatics & Data Science
- III. Discussion

Length: 1 hour

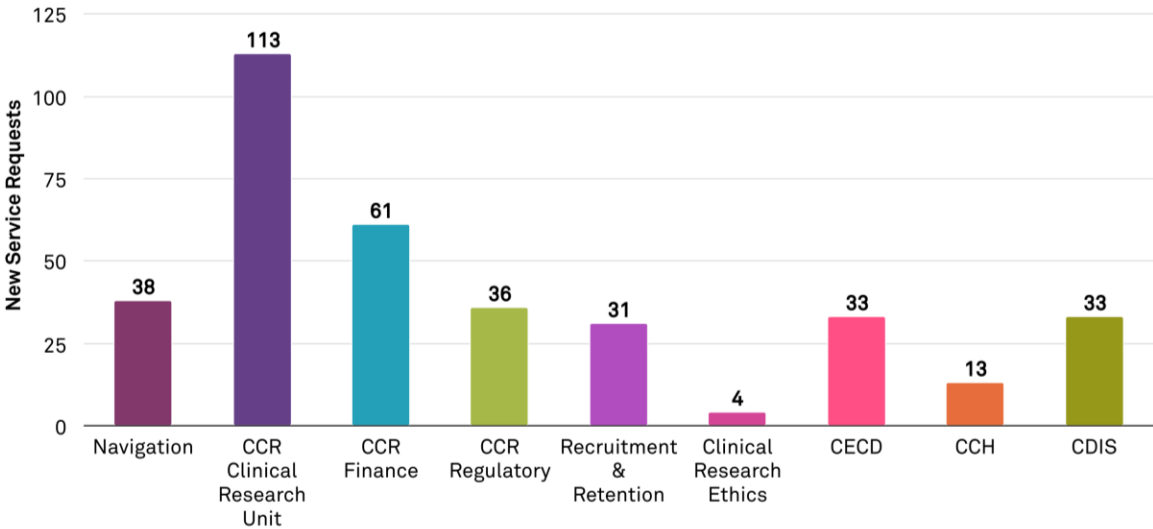
FY26 at-a-Glance

NUCATS Faculty & Staff funded by the UM1, K12, or T32 Awards or Institutional Support



111 Personnel across 21 Departments or Units

Requests for NUCATS Service, Consultations, & Resources

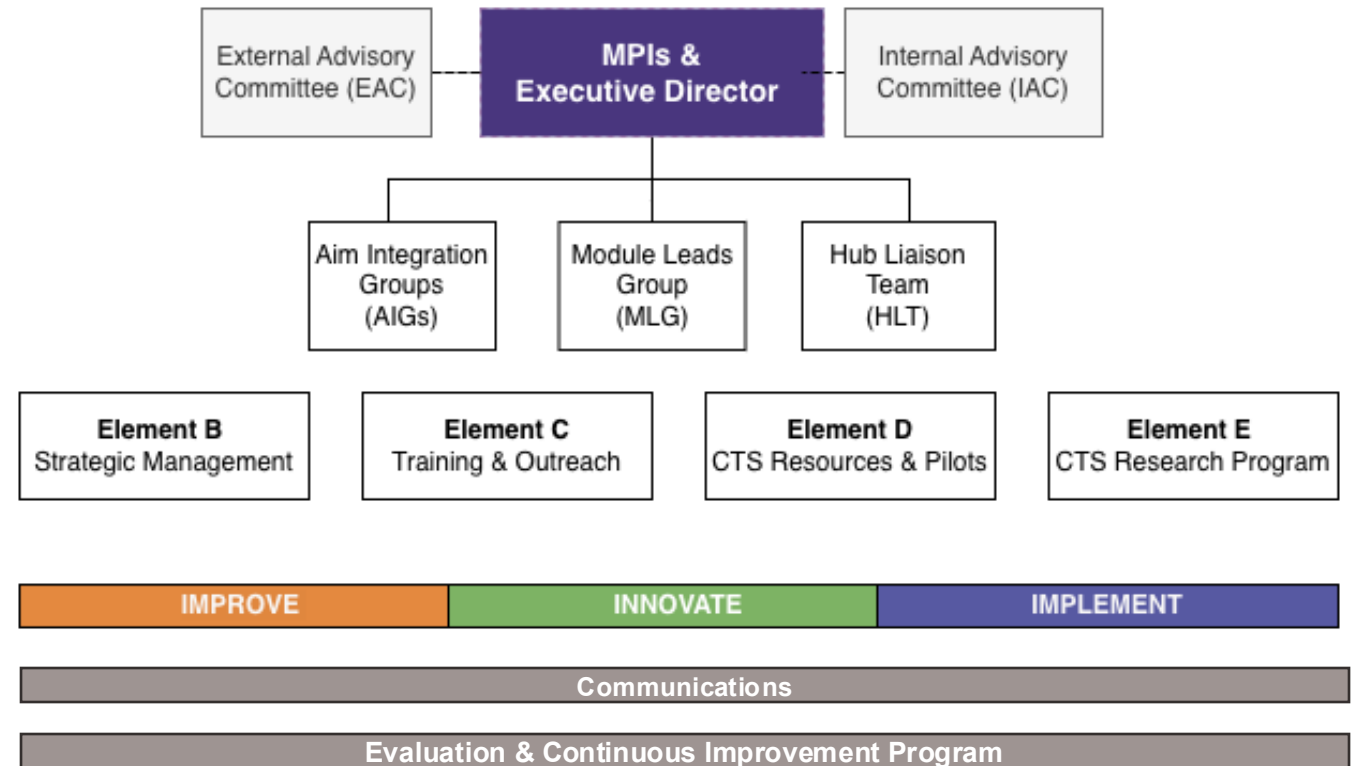
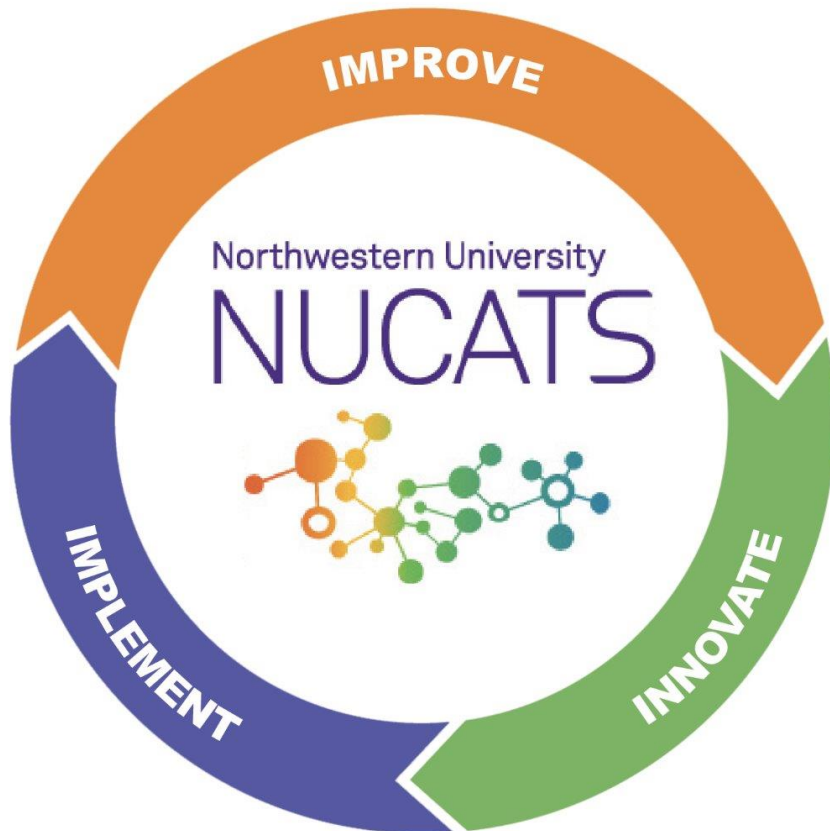


Northwestern University
NUCATS
Clinical and Translational Sciences Institute

National Center for Advancing
Translational Sciences
Site Visit | May 5-6, 2026

Hub Overview

NUCATS UM1 Organization



NUCATS

Module D3 update: Informatics & Data Science

September 2, 2026

Module D3: Informatics & Data Science

Innovating Informatics Methods & Tools

Building and Enhancing Data Science and Informatics Capacity

Focused on fostering collaboration, capacity building, and innovations in informatics and data science to advance clinical and translational research across Northwestern, its clinical partners, and the broader community.

Module D3 Co-Leads



Kristi Holmes, PhD



Faraz S. Ahmad, MD, MS

Major Accomplishments over FY2026 (Grant Year 2/7)

Building the people, platforms, and practices for a modern translational data science ecosystem

EXPANDED EXPERTISE & LEADERSHIP

Grew the core D3 team

- New co-lead with clinical and AI expertise.
- Added generative AI and bioinformatics expertise.
- Infrastructure for tracking, delivery, and communication

CLINICAL DATA INFRASTRUCTURE

- Advanced enterprise FHIR, OMOP, and Leaf/Databricks infrastructure for scalable, self-service cohort discovery.
- Refreshed i2b2: 11.7m patients, 2.4b observations; supports grant applications, recruitment, real-world data, education

COMMUNITY & COLLABORATION

- Launched Informatics & Data Science Collaborative (209 members).
- Expanded Chicagoland CTSA and national informatics partnerships.
- Increased visibility through events, seminars, and collaborative networks.

WORKFORCE DEVELOPMENT

- Delivered 16 DataLab training programs and extensive consultations.
- Trained 300+ researchers in data management, open science, and NIH compliance.
- Expanded education in research integrity, AI ethics, and RCR.

OPEN SCIENCE & RESPONSIBLE AI

- Led national Make Data Count initiatives on data recognition and evaluation.
- Advanced InvenioRDM, FAIR repositories, and NIH data-sharing infrastructure.
- Produced influential scholarship and leadership in responsible AI and research integrity.

AI & INFORMATICS INNOVATION

- Deployed adaptive randomization in clinical research.
- Piloted AI-enabled approaches for consent comprehension and participant engagement.

Self-Service Cohort Discovery

Current Status



Evaluation of Leaf



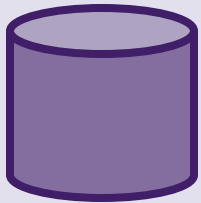
2013

Initial i2b2
deployment



2015

ACT Wave 2
Site



~11.7 million patients
~2.4 billion observations



~800

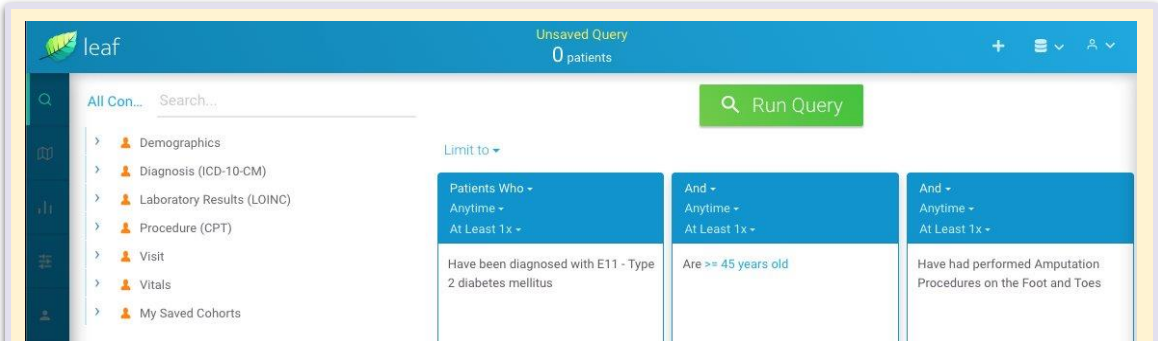
Historical
i2b2 users

30-40 active per month



Use Cases

- Grant applications
- Recruitment
- Real-world data studies
- Education



Streamlined User Experience

Targeted to feasibility queries

Data model agnostic

Includes OMOP and PCORnet as well as bespoke data structures

Extensible

Framework for customizable extensions for Leaf community

Data Infrastructure Modernization

NMEDW Cloud Migration & Enterprise-Wide OMOP

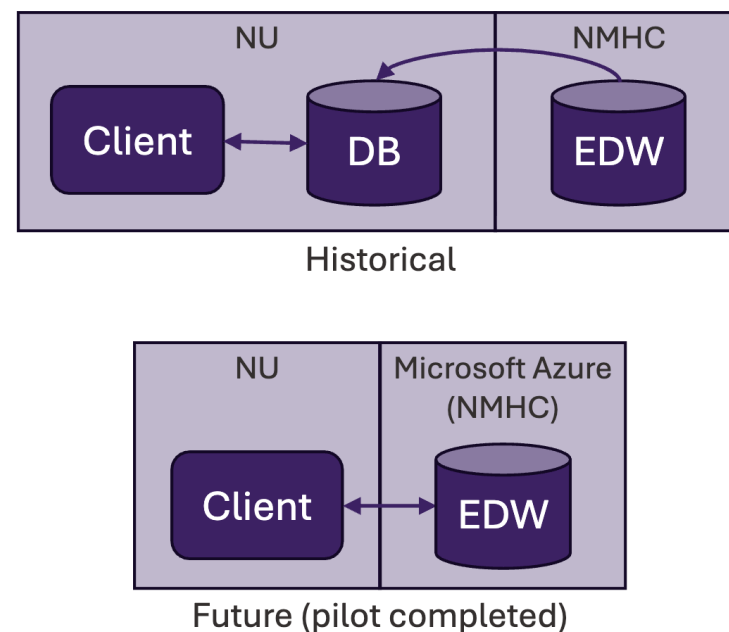
↑ Scalability

- Respond to increased demand

↓ Reduced data transfers

- More frequent data updates
- Cost savings
- Improved security controls

More timely information available
to support researchers

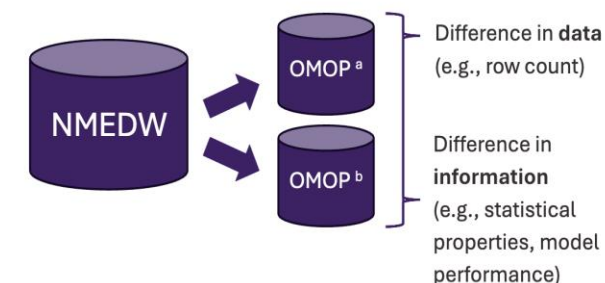


OMOP Common Data Model

The Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM) is an open community data standard, designed to standardize the structure and content of observational data and to enable efficient analyses that can produce reliable evidence.

OMOP live in EDW

- FHIR → OMOP data conversion
- Data quality assessment ongoing
- Framework to detect information loss
- Planned connection to Leaf through the NMEDW cloud migration



FY2027 Updates: From Building Capabilities to Scaling Impact



D3.1 — Innovating Informatics Methods & Tools

- a — Consent Form Transparency
- b — Cohort Discovery Tools
- c — OMOP Tooling



D3.2 — Fostering Research Integrity & Open Science

- a — Advance Open Science
- b — Enhance Research Integrity
- c — FAIR Practices and Research Data Support



D3.3 — Disseminating & Implementing Collaborative Informatics

- a — IS Evaluation Tools
- b — Translational Software Dissemination Framework
- c — Local/Regional/National Collaboration

D3.1 — Innovating Informatics Methods & Tools

Regulatory Science & Consent Form Transparency

Mohammad Hosseini

- Ongoing work with Juan Espinoza and Greg Zapatoczni from Lurie Children's and Jeff Kim of U Chicago on a paper titled "FDA Review Timeliness, Regulatory Capacity, and Postmarket Reporting During the Rise of AI-Enabled Medical Devices". The paper is currently under review in JMIR Public Health and Surveillance.
- Ongoing work with Anju Peters on a project focused on improving consent comprehension for older adults. IRB approval in June, recruiting subjects for preliminary analysis; will use pilot data to apply for full funding
- ✓ Published: Hosseini M, Eisenman D, Riddle J, Pyle S, Peters A, Cobb N, Holmes K. Guidelines needed for the use of AI in the preparation or review of IRB, IBC, and IACUC applications. Account Res. 2026 Jul;33(5):2612564. doi: 10.1080/08989621.2025.2612564. Epub 2026 Jan 8. Erratum in: Account Res. 2026 Jul;33(5):2652473. doi: 10.1080/08989621.2026.2652473. PMID: 41507141; PMCID: PMC12823195.

✓ Completed → In progress

D3.1 — Innovating Informatics Methods & Tools

Cohort Discovery Tools

Luke Rasmussen & Will Thompson | 3 of 4 milestones complete

- ✓ Leaf extended with Databricks adapter; proof-of-concept connection to NM Databricks
- ✓ Concept hierarchies built: diagnoses (ICD10), vitals, encounters, demographics
- ✓ Production Entra ID configuration in place
- Concept hierarchies underway: medications (RxNorm), procedures (CPT), labs (LOINC)

OMOP Tooling

Dan Schneider & Prasanth Nannapaneni | 1 live, 2 in progress

- ✓ FHIR → OMOP conversion live in the Enterprise Data Warehouse
- Data quality assessment ongoing with the EDW team
- Leaf connected to dev; staging and regression testing next

✓ Completed → In progress

D3.2 — Fostering Research Integrity & Open Science

Monthly Training Series

Galter Library

Developed two interactive one-hour monthly trainings through Galter Library (includes JIT guidance):

- ✓ Ethics of Authorship in Medical Research and Publications
- ✓ Ethics of Using AI in Medical Research and Publications
- Monthly delivery of content

NNCI Collaboration

AI Ethics Committee co-chair

- Ongoing collaboration with Northwestern Network for Collaborative Intelligence (NNCI) as the co-chair of “AI Ethics Committee” with Kelly Michelson
- Precision Education Governance Committee – support for participatory governance: decision making and policy development
- Various project collaborations and invited speakers (Sean Mooney, Susan Gregurick, etc.)

Scholarship & Policy Leadership

Research integrity

- ✓ Scholarship on AI disclosure, synthetic data, biosecurity risks of AI, and ethical research practices when AI is used.
- ✓ Frameworks for ethical authorship and accountability in collaborative science.
- ✓ Strengthened local and national capacity in research integrity through publications, education, and translational research collaborations.
- ✓ Thought leadership in the development of misconduct policies that were adopted by HHS Office of Research Integrity.

✓ Completed → In progress

- U.S. Department of Health and Human Services, Office of Research Integrity. (2026, May 15). *ORI's guidance on generative artificial intelligence released*.
- NIH & HHS Office of Research Integrity. (2026, May 14). *Helpful reminders to ensure integrity of NIH-supported research when using artificial intelligence*. NIH Extramural Nexus.

D3.2 — Fostering Research Integrity & Open Science

FAIR Initiatives

Data assessment & interoperability

- Data evaluation via NIH S-Index & Make Data Count (paper).
- Support for collaborative teams, including CRediT resources and forthcoming update.
- Collaborative Interoperability work with NIH.
- Coalition of Resilient Research Data Infrastructure (CRRDI); NAS new Resilient new Resilient and Enduring Data Infrastructure for Biomedical Research and Innovation (REDI).

FAIR Infrastructure

August 2026

- ✓ InvenioRDM software release (August 2026). Release notes: <https://inveniordm.docs.cern.ch/releases/v14/version-v14.0/>
- Prism migration to v14.0 underway.
- Strong growth in software adoptions in the US; strong use by NIH projects.
- Indexing NIH-funded data in Prism in the NIH Dataset Catalog.

Scholarship

Open science & scholarly communication

- ✓ Examined the opportunities and challenges generative AI presents for open science, transparency, and reproducibility.
- ✓ Promoted more transparent and equitable scholarly communication through CRediT and publishing best practices.
- ✓ Advanced responsible, sustainable, and accessible approaches to knowledge sharing and AI-enabled research.

✓ Completed → In progress

D3.3 — Disseminating & Implementing Collaborative Informatics

Project: Adaptive Randomization

IS Evaluation Tools

- ✓ Schauer JM, Broxton MO, Rasmussen LV, Swann G, Newcomb ME, Ciolino JD. Workflows to automate covariate-adaptive randomization in REDCap via data entry triggers. JAMIA Open. 2025 Oct 1;8(5):ooaf110. doi: 10.1093/jamiaopen/ooaf110. PMID: 41041624; PMCID: PMC12486239.
- IS Evaluation Tools: Schauer's adaptive randomization used in the all2GETHER study; adapted for Liver Cirrhosis Network & ASPIRIN studies

Local / Regional / National Collaborations

Several active fronts

- “Informatics and Data Science Collaborative” launched (178 members, several events)
- UChicago & UIC CTSA informatics team collaborations
- CTSA Biostatistics, Informatics, and Data Science leadership team
- ✓ Luo's CRITICAL project made publicly available (NU, WashU, UAB, Tufts, MIT); collaboration with UIC on training grant proposal (NLM T15 – pending NOFO)

✓ Completed → In progress

New Directions Under D3.1:



Innovating Informatics Methods & Tools

- Registry & Computational Phenotyping Tools
- Model Governance
- Support for Rapid-Cycle Pragmatic AI Trials



From Building Capabilities to Scaling Impact

- Ethics and Research Integrity
- Informatics and Data Science Capacity Building
- Translational Software & Data Dissemination Framework

Consultation

Training

Collaboration

Innovation

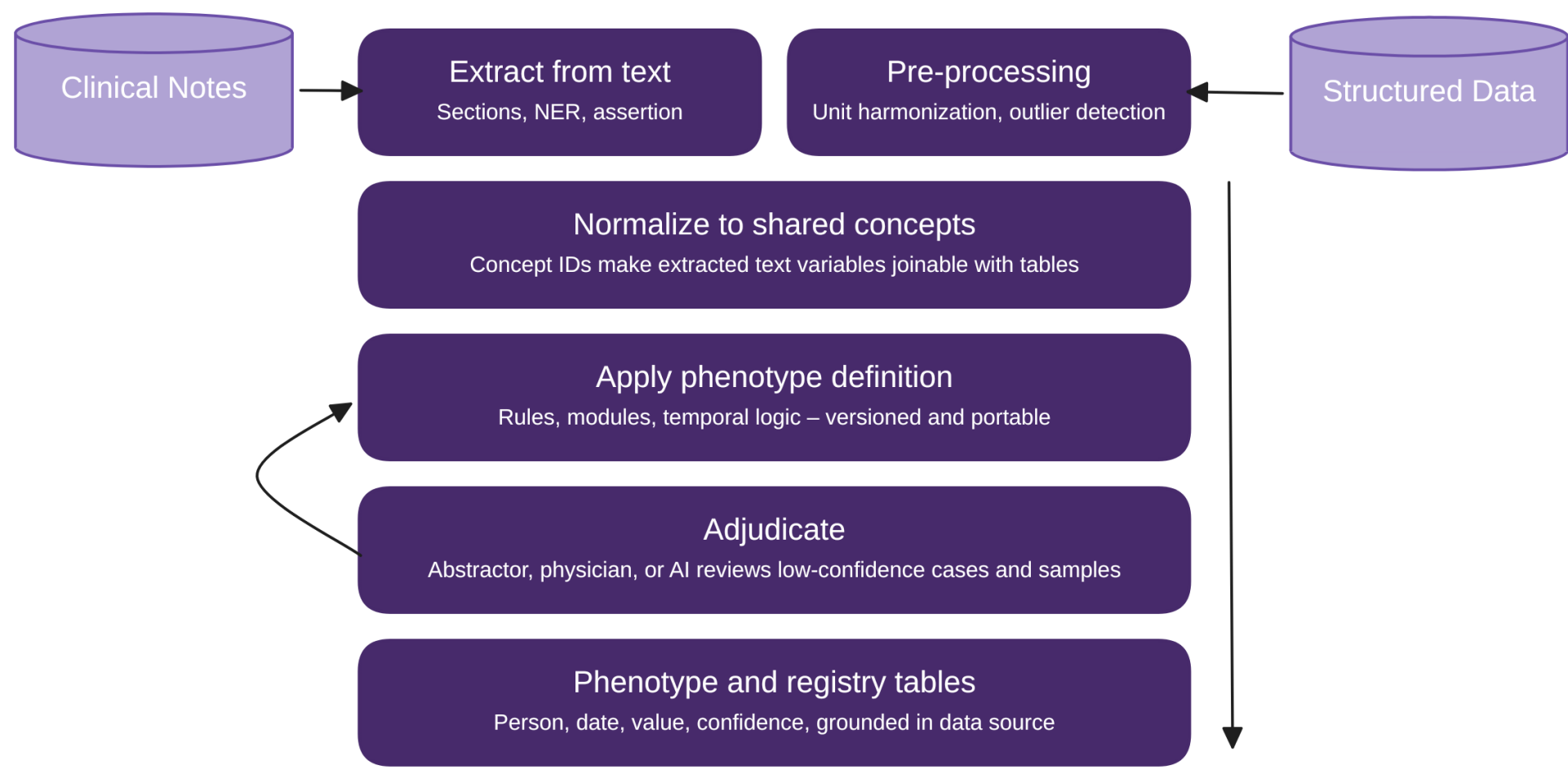
Dissemination



Innovating Informatics Methods & Tools

Registry and Computational Phenotyping Tools

Develop innovative tools using NLP and AI to dramatically reduce the time required to create computational phenotypes for registries and pragmatic clinical trials





Innovating Informatics Methods & Tools

MLflow for Model Governance

Develop tools for model versioning, evaluation, and monitoring using Mlflow in Databricks

MLflow: AI Engineering Platform for LLMs and Agents

MLflow is the largest open source **AI engineering platform for agents and LLMs**. MLflow enables teams of all sizes to debug, evaluate, **monitor**, and optimize production-quality AI applications, while controlling costs and managing access to models and data. With over 30 million monthly downloads, thousands of organizations rely on MLflow each day to ship AI to production with confidence.

MLflow's comprehensive feature set for agents and LLM applications includes production-grade **observability**, **evaluation**, **prompt management**, an **AI Gateway** for managing costs and model access, and much more.



Open Source

Join thousands of teams building agents and LLM applications with MLflow - with 26K+ GitHub Stars and 30M+ monthly downloads. As part of the Linux Foundation, MLflow ensures your AI infrastructure remains open and vendor-neutral.



OpenTelemetry

MLflow Tracing is fully compatible with OpenTelemetry with native support for GenAI Semantic Conventions, making it free from vendor lock-in and easy to integrate with your existing observability stack.



All-in-one Platform

Manage the complete AI development journey from prototype to production. Track prompts, evaluate quality, deploy AI agents, and monitor performance in one platform.



Complete Observability

See inside every AI decision with comprehensive tracing that captures prompts, retrievals, tool calls, and LLM responses. Debug complex workflows with confidence.



Evaluation & Monitoring

Stop manual testing with LLM judges and custom metrics. Systematically evaluate every change to ensure consistent improvements in your AI applications.



Framework Integration

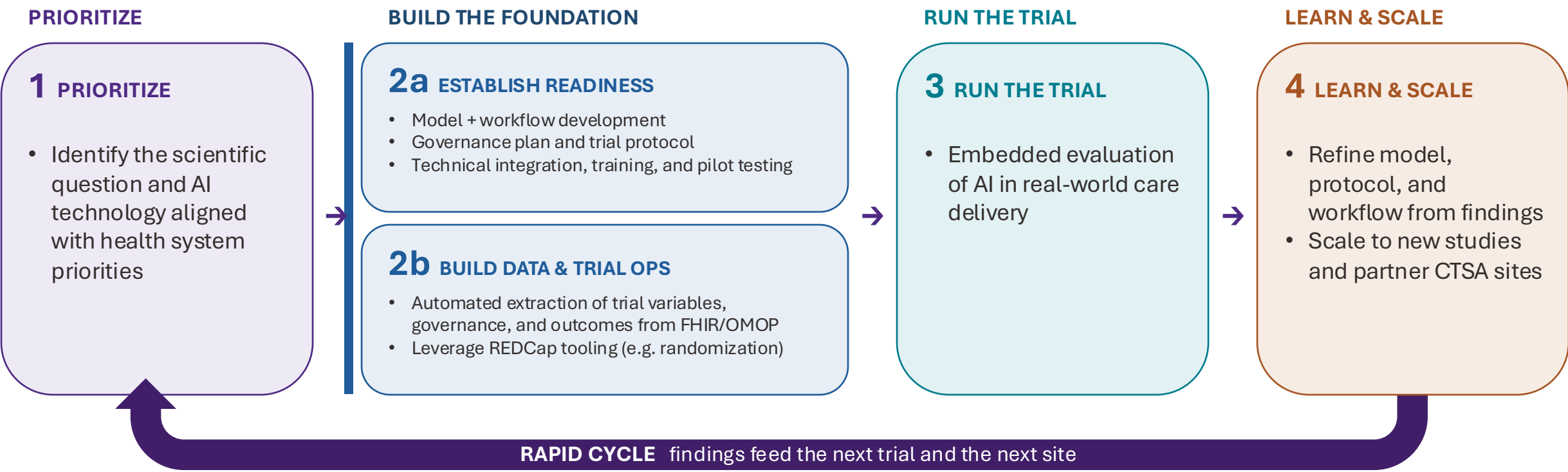
Use any agent framework or LLM provider. With 100+ integrations and extensible APIs, MLflow adapts to your tech stack, not the other way around.



- Created and open-sourced by Databricks, under Apache 2 license
- MLflow is already available to EDW Power Users through the EDW Databricks environment
- Provides tools for model and data versioning, reproducibility, evaluation, and monitoring

Innovating Informatics Methods & Tools Toward a Rapid-Cycle Pragmatic AI Trial Platform

Build reusable infrastructure to identify, launch, evaluate, and scale pragmatic hybrid trials of AI in the health system



NCATS CTSA CCIA
UG3 / UH3 PAR-25-296

- Infrastructure for rapid, robust, and ongoing evaluation of emerging AI/ML tools across health systems

- Multi-site partnership in discussion with University of Wisconsin and other sites



From Building Capabilities to Scaling Impact Coordinate Responsible Ethics and Research Integrity for Translation

Streamlined and targeted support from study design and conduct through dissemination, implementation, and policy.

01

EDUCATE

Research Ethics & Integrity Training & Consultation

- Develop and deliver programs on research integrity, authorship, and attribution, and the ethical, responsible use of AI and data science
- **Includes** transparency, reproducibility, open science practices, synthetic data, and generative AI

02

ENHANCE

Integrity Scholarship & Evidence

- Lead scholarship and evaluation examining the impact of AI on scientific integrity, research workflows, scholarly communication, and data sharing
- **Produces** publications, guidance documents, and best-practice frameworks for the translational science community

03

GUIDE

Policy, Norms & Responsible Practice

- Collaboratively develop policies and norms — including guidance on AI use, regulatory/compliance guidance, and assessment
- **Produces** tools and educational resources supporting trustworthy, equitable, and transparent practice across the research lifecycle

04

SCALE

Institutional & National Leadership

- Strengthen capacity through workshops, consultations, lectures, and collaborations
- **Promotes** open science, contributor transparency, FAIR data practices, and responsible innovation in clinical and translational research

CROSS-CUTTING THEMES

Research integrity • Authorship & attribution • Transparency • Reproducibility • Contributor roles • AI innovations • FAIR data



Building Data Science & Informatics Capacity

Exploring Coordination of a Data Science & Informatics Consultation & Collaboration Program

Accelerating translational science through accessible expertise, collaborative partnerships, and workforce development

Biomedical Data Science & AI

AI, machine learning, and analytics for discovery

Study design & data strategy / AI/ML consultation

Bioinformatics & Digital Health

Computational biology and digital health tools

Research collaboration / User experience / Data support

EXPERTISE Data Science & Informatics

Deep bench of capability
Growing capacity

Clinical & Translational Informatics

Informatics supporting research and care

Data acquisition & harmonization / REDCap support

Research Data Platforms & Engineering

Secure computing, data models, and pipelines (FAIR)

Data architecture / OMOP & FHIR modeling / Governance

(partnering with appropriate groups)

CAPACITY BUILDING

LEARN

Workshops, short courses, office hours and consultations

BUILD

Collaborative pilot projects, mentorship and workforce development

SHARE

Publications, presentations, open-source tools and shared resources

EXPECTED OUTCOMES: Stronger research teams / Increased data science adoption / Improved study quality and rigor / Accelerated translational discoveries / Enhanced open science practices / Greater AI readiness and literacy



Translational Software & Data Dissemination Framework

Maximizing Discovery, Reuse, and Impact of NUCATS Research Products

Moving NUCATS innovations beyond development to discovery, adoption, reuse, and measurable translational impact



KEY ACTIVITIES

Inventory & Understand

Product inventory and ecosystem analysis

Collaborate & Innovate

Open science & FAIR practices / Dissemination planning / Pilots

Training & Workforce Development

FAIR data workshops / Software sharing / Assessment & attribution

Scholarship & Dissemination Science

Case studies / Playbook development / CTSA publications

KEY DELIVERABLES

Dissemination Playbook / Pilot Case Studies / Peer-Reviewed Manuscript / Training Curriculum

A repeatable, evidence-based framework that makes NUCATS software, data, and AI tools discoverable, reusable, and impactful across the CTSA consortium

Benefit to Feinberg Investigators

What Each New Directions Initiative Delivers to Research Teams

Support tailored to career stage, with shared infrastructure, community, and capacity gains for Feinberg

Registry & Computational Phenotyping

EARLY CAREER Define a study cohort in days without hiring a dedicated analyst.

EVERYONE Reusable phenotypes scale registries and pragmatic trials across a portfolio.

FEINBERG Shared phenotype library reused across departments and studies.

Ethics & Research Integrity

EARLY CAREER Regulatory, IRB, and AI-disclosure guidance on authorship, research, and responsible use.

EVERYONE Policy leadership and integrity scholarship for teams and practice.

FEINBERG School-wide norms and a shared community of practice on responsible AI.

MLflow Model Governance

EARLY CAREER Versioning and monitoring already in the EDW environment – no setup cost.

EVERYONE Established infrastructure for in-silent and prospective AI model trials.

FEINBERG Model governance tooling available enterprise-wide.

Data Science & Informatics Consultation

EARLY CAREER One front door to expertise, plus workshops and mentored pilots.

EVERYONE Dedicated capacity for harmonization, OMOP/FHIR modeling, and design.

FEINBERG Central data science capacity that grows collaborative skilled teams across Feinberg.

Rapid-Cycle AI Trial Platform (long-term)

EARLY CAREER Run a first embedded AI trial on ready-made protocols and governance.

EVERYONE Multi-site path to UG3/UH3 and CCIA funding with partner CTSA hubs.

FEINBERG Reusable trial infrastructure any Feinberg team can plug into.

Translational Dissemination Framework

EARLY CAREER Citable products that also satisfy NIH Data Management & Sharing Policy.

EVERYONE Downloads, citations, and adoption as documented translational impact.

FEINBERG Institutional visibility as Feinberg products are disseminated and implemented.

Shared infrastructure, a common community of practice, and durable capacity across Feinberg

Questions/Discussion

Thank You!

Module D3 Data Science Stack

